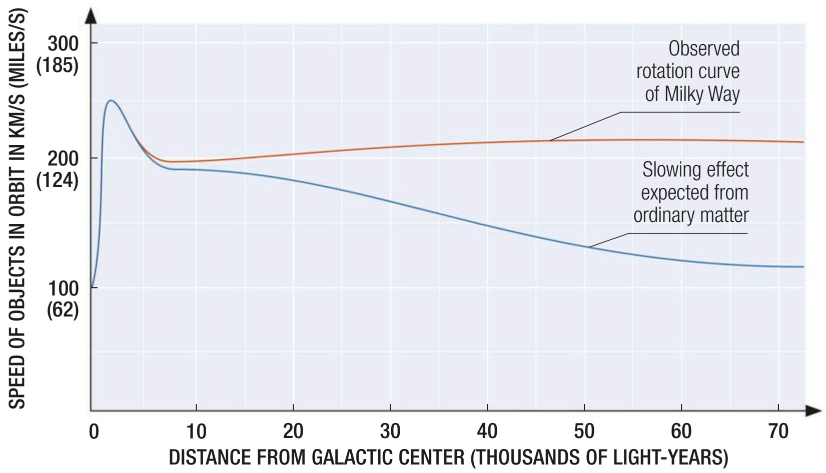


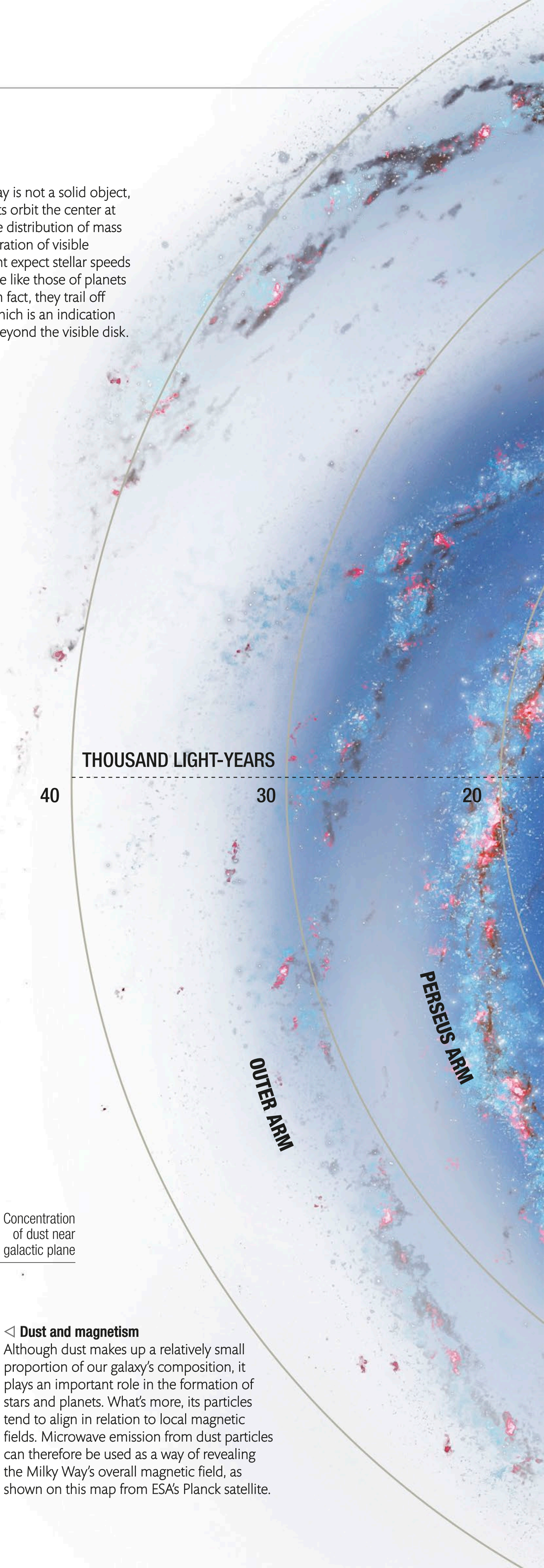
◁ **Rotation curves**
Because the Milky Way is not a solid object, stars and other objects orbit the center at different speeds. If the distribution of mass matched the concentration of visible objects, then we might expect stellar speeds to fall off with distance like those of planets in the Solar System. In fact, they trail off much more slowly, which is an indication of dark matter lying beyond the visible disk.

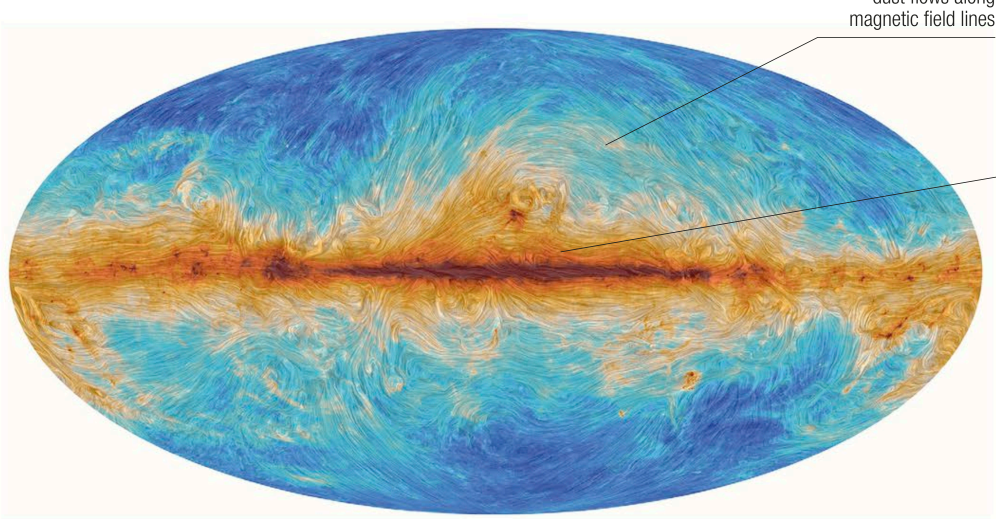
THE **MILKY WAY FROM ABOVE**

OUR GALAXY IS A VAST DISK OF STARS ABOUT 120,000 LIGHT-YEARS ACROSS. THE VAST MAJORITY OF STARS WE SEE IN OUR SKIES, HOWEVER, ARE CONFINED TO A MUCH SMALLER NEIGHBORHOOD AROUND OUR SOLAR SYSTEM.

The Milky Way contains between 100 and 400 billion stars, most of which are dwarfs with only a fraction of the Sun's mass. Nevertheless, the Galaxy's overall mass is between 1 and 4 trillion solar masses—far more than the combined mass of its stars. While much of the extra mass is accounted for by dust and gas in the galactic disk, all this normal matter is vastly outweighed by so-called dark matter (see pp.74–75).

Studies suggest that there are at least 100 billion planets orbiting the stars, and perhaps many more. Most of the visible material is in a central bulge and a flattened disk just 1,000 light-years deep, with stray stars, globular clusters, and large amounts of dark matter in the halo region.





◁ **Dust and magnetism**
Although dust makes up a relatively small proportion of our galaxy's composition, it plays an important role in the formation of stars and planets. What's more, its particles tend to align in relation to local magnetic fields. Microwave emission from dust particles can therefore be used as a way of revealing the Milky Way's overall magnetic field, as shown on this map from ESA's Planck satellite.